

Module/Pathway/Taxon Review: Alginate O-Acetylation in *Pseudomonas putida* KT2440

Review topic: `alginate_o_acetylation` — KEGG bucket `ppu00543` ("Exopolysaccharide biosynthesis") **Target taxon:** *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Module area:** other_kegg_pathway · **Candidate gene count:** 11

Summary

The alginate O-acetylation module is **COVERED and fully satisfiable** in *Pseudomonas putida* KT2440. All four canonical machinery components required to O-acetylate the alginate polymer — **AlgI** (PP_1280 / Q88ND2), **AlgJ** (PP_1279 / Q88ND3), **AlgF** (PP_1278 / Q88ND4), and **AlgX** (PP_1282 / Q88ND0) — are present in the candidate list, correctly named, and sit in the expected syntenic positions within a complete *algD* biosynthesis operon (PP_1277–PP_1288). This operon reproduces the *P. aeruginosa* gene order exactly, and the KT2440 *alg* cluster is experimentally functional (its deletion reduces biofilm-mediated drought resilience). O-acetylation is therefore biologically plausible in the target strain, not merely a homology transfer.

Two categories of candidate genes are **not** part of the alginate O-acetylation machinery and are artifacts of a coarse KEGG bucket. The four "serine O-acetyltransferase / serine acetyltransferase (EC 2.3.1.30)" entries — **PP_0228**, **cysE/PP_0840**, **PP_1110**, and **PP_3136** — catalyze the committed step of cysteine biosynthesis (L-serine + acetyl-CoA → O-acetyl-L-serine), which is mechanistically and structurally unrelated to polymer O-acetylation. They were pooled into `ppu00543` by a generic "O-acetyltransferase" grouping and represent **over-propagated annotation**. **PP_2124** ("glycosyl transferase") lies outside the *alg* operon and is an ambiguous, non-specific glycosyltransferase that should not be counted toward alginate biosynthesis. Together, five of eleven candidates (45%) are off-pathway or ambiguous.

The single most important open question is that **no direct biochemical demonstration of alginate O-acetylation exists in *P. putida* KT2440**. The COVERED assignment rests on genomic completeness (a full syntenic *algI/J/F/X* operon), a functional *alg* cluster with a documented phenotype, and strong homology transfer from *P. aeruginosa*, where isogenic knockouts and catalytic-triad mutants directly establish each gene's role. A strain-specific FT-IR or NMR acetylation assay on purified KT2440 alginate — or an isogenic *algX/algI* knockout — would convert this from a strong inference into a direct demonstration.

Key Findings

F001 — The AlgI/AlgJ/AlgF/AlgX O-acetylation machinery is present and satisfiable in KT2440

All four canonical O-acetylation genes appear in the KT2440 candidate list mapped to the *alg* operon: **algF** (PP_1278 / Q88ND4), **algJ** (PP_1279 / Q88ND3), **algI** (PP_1280 / Q88ND2), and **algX** (PP_1282 / Q88ND0). The functional roles are anchored by direct genetics in *P. aeruginosa*: isogenic knockouts of *algI*, *algJ*, and *algF* each yield O-acetylation-negative alginate ([PMID: 12003941](#), [PMID: 8636017](#)), and site-specific mutation of the AlgX Ser-His-Asp catalytic triad abolishes acetylation in vivo ([PMID: 23779107](#)). AlgI, AlgJ, and AlgF form a complex that acts together with AlgX ([PMID: 31900562](#)). Because the KT2440 genes are syntenic orthologs in a complete operon, transfer of this model is **strong**.

F002 — The four EC 2.3.1.30 serine O-acetyltransferases are over-propagated cysteine-biosynthesis annotations

PP_0228 (Q88RA5), *cysE*/PP_0840 (Q88PL0), PP_1110 (Q88NU4), and PP_3136 (Q88I65) are all annotated serine O-acetyltransferase / serine acetyltransferase (EC 2.3.1.30). EC 2.3.1.30 catalyzes L-serine + acetyl-CoA → O-acetyl-L-serine, the committed step of cysteine biosynthesis (KEGG *ppu00920*, sulfur/cysteine metabolism). This reaction is mechanistically and structurally unrelated to alginate polymer O-acetylation, which uses the SGNH-hydrolase and MBOAT-like AlgI/J/F/X proteins — **none of which carry EC 2.3.1.30**. Their inclusion in `ppu00543` reflects a generic "O-acetyltransferase" EC/GO grouping, not alginate involvement. *cysE*/PP_0840 is the canonical cysteine-pathway enzyme; the other three are CysE paralogs (a cysteine-pathway curation question orthogonal to alginate).

F003 — KT2440 carries a functional alginate (*alg*) exopolysaccharide gene cluster

Mekureyaw et al. 2026 used KT2440 mutants "Alg" (single *alg* EPS cluster deleted) and "Q" (four EPS clusters *alg*, *bcs*, *pea*, *peb* deleted); loss of the *alg* cluster reduced biofilm-mediated drought resilience, demonstrating the KT2440 *alg* cluster is functional and expressed (PMID: 41554215). Related *P. putida* strain XMS-1 shows *algD* and *algB* are functional sodium-alginate biosynthesis genes (PMID: 41633212, PMID: 40602902). The genuine *alg* operon members are PP_1278–PP_1287 (*algF*, *algJ*, *algI*, *algX*, *alg44*, *alg8*).

F004 — KT2440 encodes the complete canonical *algD* operon in *P. aeruginosa* gene order

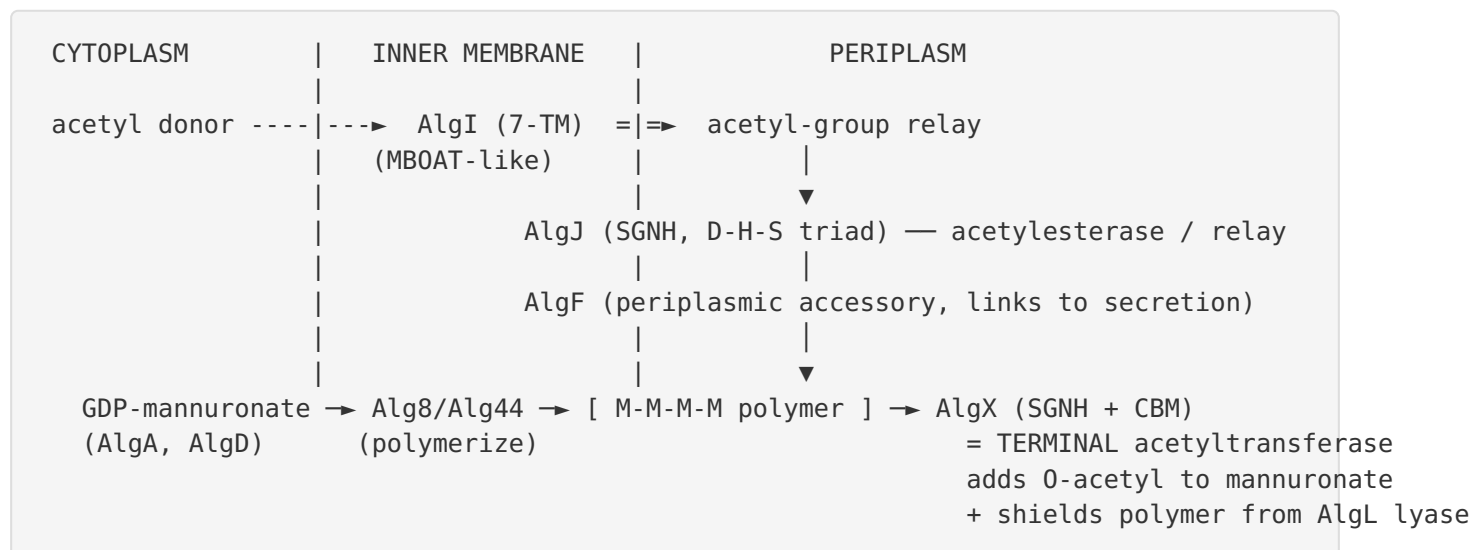
UniProt mapping (taxon 160488) of the locus range establishes the full operon: PP_1288 *algD* (GDP-mannose 6-dehydrogenase, Q88NC4), PP_1287 *alg8* (glycosyltransferase), PP_1286 *alg44*, PP_1285 *algK*, PP_1284 *algE*, PP_1283 *algG* (mannuronan C5-epimerase), PP_1282 *algX*, PP_1281 *algL* (alginate lyase), PP_1280 *algI*, PP_1279 *algJ*, PP_1278 *algF*, PP_1277 *algA*. This is the full *P. aeruginosa* *algD* operon order (*algD*-*alg8*-*alg44*-*algK*-*algE*-*algG*-*algX*-*algL*-*algI*-*algJ*-*algF*-*algA*), transcribed opposite to locus numbering. The KEGG ppu00543 candidate bucket captured only 6/12 operon genes (*algF*, *algJ*, *algI*, *algX*, *alg44*, *alg8*); *algD*, *algA*, *algK*, *algE*, *algG*, and *algL* are present in the genome but absent from the candidate list because they sit in other KEGG maps — a bucketing artifact, not a biological gap.

F005 — KT2440 O-acetylation is inferred from synteny/homology; no direct biochemical demonstration exists (open question)

Targeted PubMed searches for *P. putida*/environmental-strain alginate O-acetylation returned no strain-specific biochemical study. The functional importance of alginate O-acetylation is established in *P. aeruginosa*: FT-IR shows the O-acetyl ester band ($\sim 1250\text{ cm}^{-1}$) in mucoid biofilms and its loss in non-acetylating strains, with acetylation required for microcolony/biofilm formation (PMID: 11208804). The SGNH-hydrolase acetyltransferase paradigm (*AlgX*/*AlgJ*) generalizes to related *Pseudomonas* EPS acetylation systems (e.g., cellulose WssI in *P. fluorescens*; PMID: 37224964). KT2440 evidence remains genomic (complete syntenic *algI/J/F/X* operon) plus phenotypic (functional *alg* cluster).

Mechanistic Model / Interpretation

Alginate O-acetylation is a **post-polymerization periplasmic modification**: acetyl groups are added at the **O-2 and/or O-3 positions of D-mannuronate residues** after the polymer is synthesized and translocated across the inner membrane. Critically, loss of the acetylation genes does **not** affect polymer synthesis or C-5 epimerization — only the acetylation state changes (PMID: 8636017).



The KT2440 operon reproduces this architecture gene-for-gene. The step-to-gene mapping is unambiguous:

Step	Function	Gene	KT2440 locus	UniProt	Call
A1	Acetyl translocation across inner membrane	<i>algI</i>	PP_1280	Q88ND2	covered
A2	Periplasmic acetylerase / relay (SGNH)	<i>algJ</i>	PP_1279	Q88ND3	covered
A3	Periplasmic accessory, links to secretion	<i>algF</i>	PP_1278	Q88ND4	covered
A4	Terminal polymer O-acetyltransferase (SGNH+CBM)	<i>algX</i>	PP_1282	Q88ND0	covered

Interpretation for curation: the module is COVERED. The two SGNH proteins have distinct roles — AlgJ exhibits acetylerase activity but weak/no polymer binding, whereas AlgX binds polymannuronate length-dependently and is the terminal transferase (PMID: 25165982). The

ppu00543 bucket should be split into three curation groups: (1) alginate O-acetylation (AlgI/J/F/X), (2) alginate precursor/polymerization (Alg8, Alg44, and off-map AlgD/A/K/E/G/L), and (3) cysteine-pathway serine acetyltransferases (which must be removed to ppu00920).

Candidate Genes and Evidence

High-confidence alginate O-acetylation genes (keep / promote)

Gene	Locus	UniProt	Role	Evidence type	Transfer strength	Caveat
AlgI	PP_1280	Q88ND2	Inner-membrane acetyl translocase	Homology + synteny; <i>P. aeruginosa</i> $\Delta algI \rightarrow$ non-acetylated	Strong	EC 2.3.1.-deliberately vague; don't assign specific EC
AlgJ	PP_1279	Q88ND3	SGNH acylesterase / relay	Homology + synteny; $\Delta algJ \rightarrow$ non-acetylated	Strong	In-vitro esterase vs in-vivo relay distinction
AlgF	PP_1278	Q88ND4	Periplasmic accessory	Homology + synteny; complex member	Strong	No catalytic EC; keep descriptive
AlgX	PP_1282	Q88ND0	Terminal polymer O-acetyltransferase	Homology + synteny; triad mutants abolish acetylation	Strong	Dual role (transferase + AlgL protection)

Polymerization-context candidates (in bucket, not O-acetylation)

Alg44 — PP_1286 (Q88NC6) and **Alg8 — PP_1287 (Q88NC5, EC 2.4.-.)** are genuine *alg* operon members involved in polymerization/co-polymerase function, not acetylation. They belong in the exopolysaccharide bucket but under the **polymerization** step, not the acetylation sub-module.

Likely over-propagated / off-pathway candidates (exclude from module)

Gene	Locus	UniProt	Annotation	Verdict
PP_0228	PP_0228	Q88RA5	Serine O-acetyltransferase EC 2.3.1.30	Over-propagated — cysteine biosynthesis
cysE	PP_0840	Q88PL0	Serine O-acetyltransferase EC 2.3.1.30	Over-propagated — canonical CysE
PP_1110	PP_1110	Q88NU4	Serine O-acetyltransferase EC 2.3.1.30	Over-propagated — cysteine biosynthesis
PP_3136	PP_3136	Q88I65	Serine acetyltransferase EC 2.3.1.30	Over-propagated — cysteine biosynthesis
PP_2124	PP_2124	Q88L13	Glycosyl transferase (generic)	Ambiguous — non-operon, reassign

Evidence Base

PMID	Title (abbrev.)	How it supports / challenges findings
8636017	Identification of <i>algI/algJ</i> required for O-acetylation	Isogenic <i>algI/algJ/algF</i> knockouts → non-acetylated alginate; polymer synthesis & C-5 epimerization unaffected — defines the acetylation genes.
12003941	Mutant analysis & localization of AlgI/AlgJ/AlgF	AlgI/J/F essential for O-acetylation; localization supports periplasmic polymer-level modification.
23779107	AlgX structure & role in acetylation	Site-specific catalytic-triad mutation → non-acetylated alginate in vivo; AlgX is a required acetyltransferase.
25165982	AlgJ & AlgX SGNH topology, distinct roles	AlgX is terminal, polymer-binding transferase; AlgJ has esterase activity but weak polymer binding.
31900562	Analysis of the acetylation machinery	AlgI+AlgJ+AlgF form a complex acting with AlgX — defines the four-protein machinery.
15231808	<i>algI/algJ</i> cassette lateral transfer	AlgI/type-II-membrane cassette is a broad EPS-esterification family — supports homology transfer with caveats.
11208804	Role of alginate O-acetylation in biofilms	FT-IR O-acetyl band ($\sim 1250\text{ cm}^{-1}$); acetylation required for microcolony/biofilm formation — defines the direct assay for KT2440.
41554215	KT2440 biofilm & drought resilience	KT2440 <i>alg</i> EPS cluster deletion reduces biofilm/drought resilience — direct evidence the cluster is functional/expressed.
41633212 , 40602902	<i>P. putida</i> XMS-1 <i>algD/algB</i>	Functional alginate biosynthesis in related <i>P. putida</i> strain — supports species-level alginate production.
37224964	WssI cellulose O-acetyltransferase	SGNH-acetyltransferase paradigm generalizes across <i>Pseudomonas</i> EPS systems.

Key verbatim support: - [PMID: 12003941](#): "*the algF, algJ, and algI genes are known to be essential for the addition of O-acetyl groups to alginate.*" - [PMID: 23779107](#): "*In vivo studies reveal that site-specific mutation of these residues results in non-acetylated alginate.*" - [PMID:](#)

[31900562](#): "Three proteins, AlgI, AlgJ and AlgF have been implicated to form a complex and act together with AlgX for O-acetylation of alginate." - [PMID: 41554215](#): "biofilm-deficient mutants KT2440 Alg, with only one gene cluster for the exopolysaccharide alginate deleted ... were used." - [PMID: 11208804](#): "1,250 cm⁻¹ (C=O stretching of the O-acetyl group in alginate)."

Module and GO-Curation Recommendations

Module step	Gene(s)	Call
AlgI acetyl translocation	PP_1280	covered
AlgJ acylesterase relay	PP_1279	covered
AlgF accessory	PP_1278	covered
AlgX polymer O-acetyltransferase	PP_1282	covered
Serine O-acetyltransferases ×4 (EC 2.3.1.30)	PP_0228, PP_0840, PP_1110, PP_3136	not_expected_in_target_taxon (mis-bucketed → ppu00920 cysteine biosynthesis)
Generic glycosyltransferase	PP_2124	candidate_uncertain (reassign to correct EPS system)
Overall ppu00543 bucket boundary	—	module_needs_revision — split acetylation from precursor/polymerization/cysteine content

GO / EC notes: For AlgX use a specific "alginate/polysaccharide O-acetyltransferase activity" GO term rather than a generic acetyltransferase term; keep AlgF as a non-catalytic component (no EC). Do **not** apply EC 2.3.1.30 (serine O-acetyltransferase) to any *alg* gene. A dedicated **alginate O-acetylation module document** (algI/J/F/X, with the polymerization scaffold as context) is warranted, because the generic exopolysaccharide bucket conflates unrelated acetyltransferases. No new GO-term request appears necessary; existing alginate-acetylation terms suffice.

Genes to Promote to Full fetch-gene Review

1. **AlgX — PP_1282 (Q88ND0):** the catalytic transferase + AlgL-protection dual role; highest curation value; confirm SGNH triad and C-terminal CBM.
 2. **AlgI — PP_1280 (Q88ND2):** verify MBOAT-family seven-TM topology and membrane localization; keep EC generic.
 3. **AlgJ — PP_1279 (Q88ND3):** confirm SGNH Asp-His-Ser triad; distinguish in-vitro esterase vs in-vivo relay role.
 4. **AlgF — PP_1278 (Q88ND4):** confirm non-catalytic accessory annotation; note coupling to the secretion complex.
 5. **PP_2124 (Q88L13):** resolve which EPS system it belongs to; remove from the alginate bucket if unrelated.
 6. *(Lower priority — cysteine-pathway hygiene)* **cysE/PP_0840, PP_0228, PP_1110, PP_3136:** confirm as CysE + paralogs and strip from ppu00543.
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Limitations and Knowledge Gaps

1. **No KT2440-direct acetylation biochemistry.** The COVERED call is an inference. Every mechanistic knockout/mutant result derives from *P. aeruginosa*; transfer to KT2440 is **strong** (identical operon synteny, conserved gene names, functional cluster phenotype) but not experimentally confirmed in the target strain. This is the single most important open question.
 2. **AlgX dual-function uncertainty.** In *P. aeruginosa*, AlgX has both acetyltransferase and periplasmic scaffolding/polymer-protection roles; which dominates in KT2440 is unresolved.
 3. **Regulation and expression conditions.** KT2440 is non-mucoid relative to CF *P. aeruginosa*; whether the *algD* operon is expressed under standard laboratory conditions affects the in vivo relevance of acetylation.
 4. **Bucket incompleteness.** Six of twelve operon genes were absent from the candidate list (routed to other KEGG maps); a complete satisfiability audit of the broader EPS bucket requires pulling in the off-map genes (*algD*, *algA*, *algK*, *algE*, *algG*, *algL*).
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Proposed Follow-up Experiments / Actions

1. **Direct acetylation assay (highest value):** Purify alginate from KT2440 under alg-inducing conditions and run FT-IR for the $\sim 1250\text{ cm}^{-1}$ O-acetyl ester band and/or $^1\text{H-NMR}$ for O-2/O-3 acetyl signals — directly confirms or refutes O-acetylation ([PMID: 11208804](#)).
 2. **Isogenic knockouts:** Construct KT2440 ΔalgX and ΔalgJ ; assay acetylation loss to confirm functional transfer of the *P. aeruginosa* model.
 3. **Sequence/structure review:** Full `fetch-gene` review of PP_1282 (AlgX) and PP_1279 (AlgJ) to confirm the SGNH Ser/Asp-His-Ser catalytic triads are conserved.
 4. **Bucket curation:** Split `ppu00543` into acetylation, polymerization/precursor, and cysteine subgroups; reassign the four EC 2.3.1.30 genes to ppu00920; flag PP_2124 as `candidate_uncertain`.
 5. **Expression check:** Query KT2440 transcriptomic/proteomic datasets to determine whether the *algD* operon is expressed under relevant (desiccation/biofilm) conditions.
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Bottom line for curation

Mark **alginate_o_acetylation COVERED** in *P. putida* KT2440: AlgI (PP_1280), AlgJ (PP_1279), AlgF (PP_1278), and AlgX (PP_1282) are all present, correctly named, and syntenic within a complete, experimentally functional *algD* operon. Reclassify the four EC 2.3.1.30 serine O-acetyltransferases (PP_0228, *cysE*/PP_0840, PP_1110, PP_3136) as cysteine biosynthesis (`not_expected_in_target_taxon` here), flag PP_2124 as uncertain, and revise the `ppu00543` bucket to separate the alginate acetylation subset from precursor/polymerization and unrelated cysteine content. The key remaining gap is the absence of any direct biochemical proof of acetylation in KT2440.
